AI PATHFINDER PROJECT REPORT

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Title: AI Pathfinding Visualizer

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1. Introduction

Artificial Intelligence is widely used in real-world applications such as Google Maps, GPS navigation systems, robotics, and video games. One of the most important concepts in AI is *pathfinding*, which helps in finding the shortest or most efficient route between two points.

This project, **AI Pathfinding Visualizer**, is an interactive application developed in Python that demonstrates how AI algorithms work in a grid-based environment. Instead of only learning the theory, this project allows users to visually see how algorithms explore different paths and finally find the optimal solution.

The main aim of the project is to make the understanding of Artificial Intelligence algorithms simple, practical, and easy to visualize.

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2. Objective of the Project

The main objectives of this project are:

- To understand how AI pathfinding algorithms work in real life
- To implement the A* (A-Star) algorithm using Python
- To create an interactive visualization where users can see how paths are calculated

- To demonstrate how AI is used in real-world navigation systems
- To build a project that combines programming with Artificial Intelligence concepts

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3. Problem Statement

In real-world situations such as road navigation, robots moving inside buildings, and game characters finding their way through obstacles, it becomes necessary to find the shortest path quickly and efficiently. Most people use navigation systems daily but do not understand how the system actually decides the best route.

This project solves the problem by creating a simple and visual platform where users can create obstacles, select starting and ending points, and observe how an AI algorithm finds the path step-by-step.

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4. Project Overview

The AI Pathfinding Visualizer is a Python-based application built using the **Pygame library**. It creates a 2-dimensional grid where each box represents a node. The user can interact with the grid using the mouse and keyboard.

The user first selects:

- A starting point
- An ending point
- Obstacles (walls)

After that, the algorithm runs and shows how it explores different nodes until it finds the shortest path between the start and end points.

This project mainly demonstrates the **A* (A-Star) Algorithm**, which is one of the most efficient pathfinding algorithms used in Artificial Intelligence.

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5. Technology Used

The technologies used in this project are simple but powerful:

- **Programming Language:** Python

- **Library Used:** Pygame
- **Concept Used:** Artificial Intelligence (Pathfinding Algorithms)
- **Development Method:** Grid-based visualization system

The entire project is implemented using Python code, which makes it easy to understand and modify.

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6. Working of the Project

The project works in a very simple and interactive way. The main logic of the project is written in the Python file.

Step-by-Step Working:

1. When the program runs, a grid of 30 × 30 boxes is created.
2. Each box in the grid is called a **Spot**, and it behaves like a node in a graph.
3. The user clicks on the grid to:
    - Select the start point
    - Select the end point
    - Create obstacles (barriers)
4. When the SPACE key is pressed, the A* algorithm starts working.
5. The algorithm checks all possible paths and calculates the shortest distance.
6. Finally, the shortest path is displayed using a different color.

This process helps users clearly understand how AI algorithms make decisions.

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7. Features of the Project

This project includes several useful features:

- Interactive grid system
- User-defined start and end points

- Ability to create obstacles
- Real-time visualization of pathfinding
- Automatic shortest path detection
- Simple keyboard and mouse controls
- Clear and easy-to-understand interface

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8. Algorithm Used (A-Star Algorithm)

The main algorithm used in this project is the **A* (A-Star) Algorithm**. It is one of the most efficient search algorithms used in Artificial Intelligence.

The algorithm works by:

- Calculating the distance from the start node
- Estimating the distance to the end node
- Selecting the node with the lowest cost
- Continuing the process until the final path is found

This algorithm is commonly used in:

- Google Maps
- GPS systems
- Robotics
- Game development

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9. System Requirements

To run this project, the following requirements are needed:

Hardware Requirements

- Computer/Laptop
- Minimum 4 GB RAM
- Basic mouse and keyboard

Software Requirements

- Python installed
- Pygame library installed
- Code editor (such as VS Code or IDLE)

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10. Steps to Run the Project

The project can be run easily by following these steps:

1. Install Python
2. Install Pygame using the command:

pip install pygame

3. Open the project file
4. Run the Python program
5. The grid will open automatically
6. Use the mouse to place start, end, and barriers
7. Press SPACE to start the algorithm

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11. Testing of the Project

The project was tested in different situations to ensure it works correctly.

Functional Testing

- Checked if the shortest path is always found
- Tested obstacle handling
- Verified start and end point selection

Edge Case Testing

- No path available
- Start point equals end point
- Fully blocked grid

The project worked successfully in all these test cases.

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12. Advantages of the Project

- Easy to understand AI concepts
- Helps beginners learn algorithms visually
- Useful for students studying Artificial Intelligence
- Can be modified for advanced algorithms
- Simple and interactive user interface

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13. Future Improvements

This project can be improved further in many ways:

- Add more algorithms like Dijkstra's Algorithm
- Add weighted paths (traffic simulation)
- Add buttons instead of keyboard controls
- Add real-world map system
- Add performance comparison between algorithms

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14. Learning Outcomes

By completing this project, I learned:

- How Artificial Intelligence algorithms work
- How to implement A* algorithm in Python
- How real-time visualization works
- How to use the Pygame library
- How to combine programming with AI concepts

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15. Conclusion

The AI Pathfinding Visualizer project successfully demonstrates how Artificial Intelligence can be used to find the shortest path in a grid-based environment.

The project not only helps in understanding algorithms but also shows how these algorithms are used in real-world applications such as navigation systems and robotics.

This project proves that even simple AI concepts can be made easy to understand through visualization and practical implementation.